June 16, 2025

Centers for Medicare & Medicaid Services (CMS)
Office of the National Coordinator for Health Information Technology (ASTP/ONC)

VIA ELECTRONIC DELIVERY

Subject: RFI Response – Interoperability and Digital Health Advancement, File code CMS-0042-NC

Dear CMS and ASTP/ONC,

Thank you for the opportunity to comment on the advancement of digital health technologies and the continued implementation of the 21st Century Cures Act, Pub. L. 114-255 ("the Act").

Shadowbox, Inc. is a healthcare technology company addressing the reported $950 billion cost of administrative burden driven in part by a lack of interoperability across thousands of stand-alone electronic health record (EHR), practice management, social service, and ancillary services systems of record.[1] Our mission is to ensure that patient data can be securely and compliantly accessed and transferred for the provision of healthcare and social services, no matter where that data resides. Our primary target audience are healthcare and social service providers serving the "long tail" or "last mile" of care, including health systems and hospitals, independent providers, social service providers, and rural and underserved communities where traditional integration methods are often too expensive, financially risky, and technically challenging for these organizations to implement and maintain.

The Act is of particular importance to achieving our mission. Specifically, TITLE IV – DELIVERY of the Act, which directed the Department of Health and Human Services, Office of the National Coordinator (now the "ASTP/ONC"), to implement regulations requiring developers of health IT to meet certain requirements to become "certified health IT," is critical for Shadowbox and other innovative technology companies to achieve these public health objectives on behalf of healthcare and social service providers and their patients. Shadowbox's advanced and proprietary technology would be described as a "successor technology" as referenced in the Act in Section 4002 as follows:

> ''(iv) has published application programming interfaces and allows health information from such technology to be accessed, exchanged, and used without special effort through the use of application programming interfaces or ***successor technology*** or standards, as provided for under applicable law, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws;" (emphasis added)

The intent of these requirements is to standardize and organize health information to move toward the public health objective of real time data exchange across all providers. Further, the Act directed the ONC to implement regulations describing penalties for those certified health IT vendors who engage in "information blocking,"

[1] David Cutler, PhD., Harvard University, Testimony to US Senate Budget Committee, 5/8/24

which is defined in the Act as "preventing, discouraging, or interfering with the access, exchange, or use of information."

The regulations prohibiting Actors from "information blocking" are a critical component to accomplish these objectives, and from our perspective are well conceived and reflective of the spirit of the Act. However, because the applicability of these rules is limited to "certified Health IT," the impact does not yet address a vast array of health IT use-cases critical to achieving true real-time data exchange at scale.

Much progress has been made since the Act was signed into law. The major health IT vendors have invested significant resources and made measurable progress toward achieving the ONCs interoperability objectives. Billions of transactions regularly occur across the large healthcare entities deploying these technologies. However, there are three communities of opportunity for expanding the mission of interoperability with a specific focus on advancing access and equity for rural and underserved communities:

- **Specialty Practices**. There are many EHR companies that cater to specialty practices or practices that fall outside the scope of the government's incentive programs and thus remain uncertified health IT. In some instances, it is simply a matter of their limited user base not being able to cover the cost of compliance. In other instances, it is a conscious decision by that vendor to remain uncertified so they may continue to charge for access to patient data residing on their systems. Some of these vendors actively engage in information blocking as a corporate strategy.

- **Ancillary Services**. True interoperability requires all health IT vendors who host systems of record containing patient data to be subject to information blocking rules. Because the incentive programs are focused around EHR adoption and utilization by a provider, there is limited incentive for the vendors providing systems of record for Ancillary Service providers such as laboratory/radiology information systems, revenue cycle management companies, payers, and others to seek certification or refrain from information blocking. It is equally important for these Ancillary Service providers and their suite of vendors to receive and exchange information to ensure the rapid and clinically accurate delivery of services while reducing the friction endured through the revenue cycle management process.

- **Social Services**. The most significant opportunity from a health access perspective, is the fragmented, unstructured, uncoded, and undercapitalized segment of IT systems serving the social services industry. While states such as California have enacted legislation mandating real-time data exchange throughout the continuum of care, there are little to no incentives for these "health" IT vendors to become certified or refrain from information blocking.

The result of these organizations falling outside of the structure and controls provided by being certified health IT is that the Act creates a two-tiered system whereby the large and well-capitalized organizations are able to provide one level of quality and care, while the smaller, underserved, rural, specialty, and social services organizations are unable to meet these critical public health objectives.

With this in mind, the following recommendations are organized to respond to the specific questions outlined in the **CMS-0042-NC** request for information.

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1. Access Gaps Beyond Certified Health IT (PC-2, PC-8, PR-1, PR-3, TD-1)

While ONC's certified health IT program has made significant strides, the narrow application of information blocking rules to only certified systems excludes vast sectors of healthcare that house patient data—especially in behavioral health, ancillary and social services.

Recommendation:
Expand the enforcement of information blocking rules to non-certified health IT vendors whose systems house patient data derived from or reimbursed by federal healthcare programs (Medicare, Medicaid, VA, DoD). Without this expansion, patients in underserved and rural settings continue to face fragmented access and poorer health outcomes due to information blocking by non-certified vendors. Given the breadth of providers housing data derived or reimbursed by federal healthcare programs, the proposed requirement would likely capture the vast majority of currently non-certified health IT.

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2. Certification Incentives for Specialty and Social Services (PR-2, TD-2)
Vendors serving specialty, ancillary, and social service providers frequently remain uncertified due to cost or strategic choice. This limits the potential for digital health products that can leverage data from these sectors.

Recommendation:

- Create a funding mechanism to support certification for vendors serving these critical sectors.
- Establish contracting requirements that mandate certification or attestation to information blocking compliance for federal health contractors and subcontractors.
- Develop a searchable registry of non-certified vendors holding federally reimbursed patient data.

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3. Tracking, Reporting, and Enforcing Information Blocking (PC-5, PC-13, TD-12, PR-4)
The ASTP/ONC Information Blocking Portal is a powerful tool, but its scope and effectiveness is limited. Currently, aggregated reports of "Non-CHPL-listed Health IT Developer" (defined by ASTP/ONC as an entity not listed in the "Developer" column of the Certified Health IT Product List but appears from the facts and allegations as presented by the claimant to develop or offer health information technology as the term is defined in 42 U.S.C. 300jj(5)) is neither a deterrent nor incentive for those non-certified vendors to refrain from information blocking.

Recommendation:

- Expand ONCs public reporting of information blocking complaints involving "Non-CHPL-listed Health IT Developer" by making accessible a searchable database of those complaints and a status report regarding ONC's determination as to whether or not the actor is a developer of health information technology as defined in 42 U.S.C. 300jj(5).
- Direct ONC to notify named vendors, share certification resources, and track/report those vendors who become certified over time.
- Explicitly enable third parties to submit reports to the Office of Civil Rights on behalf of patients whose records reside in health IT systems (certified and non-certified) when a health IT vendor intervenes to prevent the compliant delivery of that patient's health data.

- Extend the Office for Civil Rights reporting portal to include a category for reporting vendors who engage in retaliatory practices against providers attempting compliant data exchange.
- Proactively enforce information blocking prohibitions in the Act prohibiting unreasonable contract terms in health IT vendor's Terms of Services (TOS) Agreements that prevent, discourage, or interfere with the access, exchange, or use of information for the provision of healthcare services. An example of such a term, which is seen across many TOS agreements of both certified and non-certified health IT vendors, is as follows:

> "Shall not … allow any User, employee, contractor or agent to access the Services, with, for example, any automated or other process such as robotic process automation, screen scraping/data scraping/web scraping, by using robots, web-crawlers, spiders or any other sort of bot or tool, for the purpose of extracting data, monitoring availability, performance, functionality, or for any other benchmarking or competitive purpose…" (*Retrieved on June 12, 2025*)

Such a term should be updated to include the words, "Unless otherwise authorized by the User… User Shall not…"

Users should never be discouraged from utilizing a third-party IT solution or be threatened with discontinuation of services by an IT vendor when a third-party solution is authorized by that practice or User. Without enforcement, rapidly advancing technologies like Agentic AI and Robotic Process Automations may be thwarted. These tools meet public health objectives and the letter of the Act by improving access to patient data, enhancing workflows, reducing error rates, and ultimately improving patient care.

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4. Driving Digital Adoption Through Education and Demand (PC-5, PC-6, PR-1, PR-4)
Providers often lack clarity about the benefits of certified health IT and compliant vendors. This allows low-quality systems to persist in the market.

Recommendation:

- Fund education programs targeting providers, emphasizing the value of certified health IT.
- Launch patient- and provider-focused communications campaigns on how to identify and use certified health IT apps.
- Encourage CMS to play a role in vetting or signaling preferred digital health apps based on objective interoperability criteria.
- Publish an Office of Inspector General Bulletin on the effect of information blocking by non-certified vendors.

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5. Leveraging TEFCA (TD-6, PC-10)
While TEFCA provides a framework, it must also accommodate participation from smaller vendors and advance utilization of successor technology.

Recommendation:

- Identify and document the need for "successor technology" directed at achieving scalable, secure, compliant data access and collection that minimizes the cost and technical burden on the "long tail" of health and social service providers
- Pilot a TEFCA-middleware solution that utilizes "successor technology" to achieve this goal through utilization of a bi-directional front-end integration solution for web-hosted EHR applications.
- Refrain from codifying standards solely based on API technologies.

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6. Key Use Cases for Future Innovation (PC-12, TD-13)
Digital tools should support provider and patient workflows such as real-time scheduling, automated intake forms, and binding cost estimates.

Recommendation:

- Identify and implement key successor technologies that utilize front-end bi-directional integration to web-hosted EHR applications enabling rapid, scalable, and low-cost data access and exchange that do not require back-end integration to EHR vendor systems.
- Identify and implement key successor technologies that enable users to select from a menu of best-in-class applications which may or may not be available as a feature/function within the EHR or system of record used by the health or social service provider. Examples of such tools include clinical decision support tools, AI-enabled structuring and codification of unstructured data, payer prior authorization initiation/automation tools, patient fee estimate, cash pay/co-pay/deductible collection at time of order, etc.
- Create a policy and procedure for authenticating agentic AI and other automation solutions when compliantly and securely encountering Multi-Factor Authentication authorized by a health or social service provider.

CMS and ASTP/ONC have made commendable progress toward the public health objective of universal real time data exchange. To truly realize a patient-centric, equitable, interoperable system, we must now expand enforcement, certification support, and data access to include the long tail of the healthcare and social service ecosystems.

I appreciate the opportunity to submit these comments and am available for further discussion.

Sincerely,

Gregory A. Stein, Founder & CEO
Shadowbox, Inc.